MILITARY SPECIFICATION

SYSTEM, AUTOMATIC, WATER ACTIVATED PARACHUTE RELEASE;

DETAIL SPECIFICATION FOR

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 This specification established the general requirements for the design, construction, installation and performance of an automatic parachute release system for aircrewmen.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein. This specification shall take precedence over subordinate specifications listed below on any individual point of conflict.

SPECIFICATIONS

Federal

ZZ-R-765	Rubber, Silicone
QQ-C-320	Chromium Plating (Electrodepositioned)

Military

MIL-I-10	Insulating Materials, Electrical, Ceramic
MIL-M-14	Molding Plastics and Moulded Plastic Parts, Thermosetting
MIL-P-997	Plastic Material, Laminated, Thermosetting, Elec Insulation
MIL-S-5002	Surface Treatments and Metallic Coatings for Metal Surfaces of Weapon Systems
MIL-E-5400	
MIL-T-5422	Testing, Environmental, Airborne Electronic and Associated Equipment
MIL-C-5541	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-C-6021	Casting, Classification and Inspection of
MIL-H-6088	Heat Treatment of Aluminum Alloys
MIL-H-6875	Heat Treatment of Steels (Aircraft Practice), Process for
MIL-F-7179	Finishes and Coatings; General Specification for Protection of Aircraft and Aircraft Parts
MIL-F-7190	Forgings, Steel, for Aircraft and Special Ordnance Applications
MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aircraft and Missiles
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys

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PUBLICATIONS

Naval Air Systems Command

SD24	General Specification for Design and Construction of Aircraft Weapon Systems

REPORTS

NAEC-ACEL

Report No. 533	Anthropometry of Naval Aviators - 1964
NAVWEPS OD 30393	Design Principles and Practices for Controlling Hazards of Electromagnetic Radiation to Ordnance (Hero Design Guide)

3. REQUIREMENTS

3.1 Selection of materials and standard parts. The selection of materials, standard parts, processes, corrosion protection, and design features significant in adequate corrosion behavior shall be in accordance with the requirements of Naval Air Systems Command Design Specification SD24.

3.1.1 Materials. Material shall conform to applicable specifications as specified herein. Materials which are not covered by government specifications, or which are not specifically described herein, shall be of the best quality,of the lightest practicable weight, suitable for the purpose intended, and shall be approved by the Government procuring activity. Particular care shall be given to close fitting parts in the choice of both materials and corrosion practices.

3.1.1.1 Metal parts. All metal parts shall be of the corrosion-resistant type or treated in a manner to render them resistant to corrosion. Unless suitably protected against electrolytic corrosion, dissimilar metals, as defined in MIL-STD-889, shall not be used in contact with each other.

3.1.1.1.1 Heat treatment. Heat treatment of aluminum parts and steel parts shall be in accordance with MIL-H-6088 and MIL-H-6875, respectively.

3.1.1.1.2 Castings. Castings used in the automatic release system shall conform with the requirements of MIL-C-6021. Aluminum alloy castings shall conform to the requirements of MIL-A-21180.

3.1.1.1.3 Steel forgings. Steel forgings used in the system shall conform to the requirements of MIL-F-7190. Critical forgings shall meet the requirements of MIL-F-7190 Grade A forgings.

3.1.1.1.4 Magnesium and Magnesium Alloys. Magnesium and magnesium alloys shall not be used.

3.1.1.2 Non-metallic components. Non-metallic components shall be designed to minimize deterioration caused by abrasion and/or exposure to sunlight, microorganisms, moisture, heat, fuel, hydraulic and lubricating oil and grease, and salt spray.

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SPECIFICATIONS

Military (Continued)

MIL-D-8634	Decal, Elastomeric Pigmented Film, for Use on Exterior Surfaces
MIL-C-8779	Color, Interior, Aircraft, Requirements for
MIL-P-15037	Plastic Sheet, Laminated, Thermosetting, Glass-Cloth
MIL-P-15047	Plastic Sheets, Laminated Thermosetting, Nylon Fabric Base
MIL-N-18307	Nomenclature and Identification for Elec Aeronautical Support
MIL-S-18471	System, Automated Aircrew Escape, Ejection Seat Type, General Specification for
MIL-A-21180	Aluminum Alloy, Castings, High Strength
MIL-D-21625	Design and Evaluation of Cartridges, for Cartridge Actuated Devices
MIL-D-23615	Design and Evaluation of Cartridge Actuated Devices
MIL-I-23659	Initiator, Electric, Design and Evaluation of
MIL-D-23890	Decalcomanias, Process for Application of
MIL-A-25165	Aircraft Emergency Escape System, Identification of
MIL-C-26074	Coating, Electroless Nickel, Requirements for
MIL-M-38510	Microcircuits, General Specification for
AD42	Naval Air Systems Quality Assurance Procedures, Requirements for Preparation of
AR49B	Aircrew Automated Escape Systems Reliability and Maintainability (R/M Program:-Requirements for)
WR62	Naval Weapons Requirements, Specifications and Standards; Use of

STANDARDS

Military

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-142	
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-471	Maintainability/Verification/Demonstration/Evaluation
MIL-STD-810	Environmental Test Methods
MIL-STD-838	Lubrication of Military Equipment
MIL-STD-882	System Safety Program for Systems and Associated Subsystems and Equipment, Requirements for
MIL-STD-889	Dissimilar Metals
MIL-STD-1385	
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities

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Protection shall be provided for those non-metallic components, particularly nylon lines, for which strength degradation associated with abrasion or exposure-induced deterioration can endanger the system user.

3.1.1.2.1 Plastic. Laminated plastic material shall conform to MIL-P-997, MIL-P-15037 or MIL-P-15047. Molded plastic material shall conform to MIL-M-14. Cotton-filled or wood-flour-filled materials shall not be used.

3.1.1.2.2 Ceramic. Ceramic insulating material shall conform to MIL-I-10, Grade L422 or higher. Ceramic used for external surfaces shall be glazed.

3.1.1.2.3 Rubber. Rubber shall conform to ZZ-R-765.

3.1.1.2.4 Potting Compounds. The selection and use of encapsulating materials (potting compounds) shall be in accordance with MIL-STD-454.

3.1.1.3 Lubrication. Lubricants and lubrication practices shall conform to the requirements of MIL-STD-838. Lubricants shall be permanent and shall function satisfactorily throughout the temperature range from -65° to +200°F. Choice of lubricants shall (a) not degrade the function of non-metallic escape system components, (b) not damage finishes adjacent to location of lubricated areas and (c) eliminate the need for relubrication by field maintenance activities.

3.1.1.4 Fungus-proof materials. To the greatest extent practicable, the materials used shall not be nutrients for fungi. If materials that are nutrients for fungi must be utilized, such materials shall be treated with a fungicidal agent approved by the government procuring activity.

3.1.2 Corrosion protection. Corrosion protective practices employed in the manufacture of the system and its components shall be in accordance with the MIL-P-7179 requirements for exterior surfaces.

3.1.2.1 Finishes. Protective coatings and finishes shall not crack, chip, or scale during normal service life, or in the herein specified extremes of atmospheric conditions. Surface treatments, coatings and finishes shall conform to MIL-S-5002, except that all aluminum and aluminum alloy parts shall be anodized or chemically surface treated in accordance with 3.1.2.2.1.

3.1.2.2 Anodizing. All aluminum and aluminum alloy parts, except those subject to wear, shall be anodized in accordance with MIL-A-8625, Type II anodic coating. Anodic coatings for all aluminum and aluminum alloy parts subject to wear shall conform to MIL-A-8625, Type III, except for parts which are expensive and would normally be reworked during overhaul. For these parts, chromium plating in accordance with QQ-C-320 shall be used.

3.1.2.2.1 Chemical surface treatment. For aluminum and aluminum alloy parts not subject to wear, abrasion or erosion, chemical conversion surface treatment in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.1.2.3 Plating. Steel parts shall be plated in accordance with MIL-C-26074A, Electroless Nickel, Class 2. The finish shall be dull, lusterless and non-reflective.

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3.1.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 and WR62.

3.1.4 Standard parts. Standard Established Reliability (ER) electronic components shall be utilized to the maximum extent possible and shall be selected in accordance with MIL-E-5400 and MIL-STD-454. With exception of any batteries submittal request for non-standard parts approval shall not be required as the electronics assembly is a non-repairable unit.

3.2 System performance. The system, defined as the modified parachute release fitting as well as the sensing and actuation package, shall:

a. Provide the capability to release the parachute under a tensile load of 600 lbs.

b. Automatically operate when totally immersed in water with a conductivity of 2500 micromhos or greater at standard conditions.

c. Operate within 2 seconds after water immersion.

d. Not affect normal or emergency procedures or capabilities including normal and emergency ingress/egress except as specified above.

e. Not operate when deluged by heavy rains or exposed to 100% humidity or other environmental conditions that may be encountered during the operation or Naval aircraft in performance of their mission.

3.3. Design considerations. The automatic parachute release system shall be designed for functional and physical interface with the aircrew and his equipment.

3.3.1 General. The requirements of MIL-E-5400 for Class 2 Category III equipment apply as requirements of this specification with exceptions and additions as stated herein.

3.3.1.1 Physical considerations. Low weight and small size shall be a primary consideration in construction. With the exception of the requirement for ease of battery and EED replacement, the electronic assembly shall be totally encapsulated and non-repairable. The unit shall be designed in such a way as to preclude any damage from battery leakage. It shall present no bodily harm or discomfort to the aircrew. Colors of components shall be compatible with MIL-C-8779. System components which leave the aircraft during an escape must have an unobtrusive color and be non-reflective, in order to reduce the likelihood that such pieces or components will assist in the apprehension of any of the crewmembers during the enemy-evasion phase of their escape.

3.3.1.2 Energizing. The automatic release system shall be energized during water immersion.

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3.3.1.3 Interface with the aircrew and aircrew equipment. Human factors general design considerations shall be in accordance with MIL-STD-1472. Crew accommodation including the installation of the system into existing aircraft escape systems, shall not adversely affect comfort, cockpit compatibility, or the ability to accommodate 3rd through 98th percentile aircrewmen in accordance with MIL-S-18471 and NAEC-ACEL Report No. 533.

a. All components shall be neatly integrated into the existing aircraft escape system and shall comply with the applicable requirements of MIL-S-18471.

b. The system shall be compatible for installation within close proximity of an aircrewman's head with no outgassing and minimal noise upon actuation.

c. System installation upon the aircrew equipment shall not have an adverse effect upon functional reach to normal or emergency controls and shall not provide an area of discomfort to the aircrew.

d. System installation upon the aircrew equipment shall not effect normal ingress/egress or emergency escape procedures or capabilities.

e. The system shall be totally installed on the parachute assembly and therefore have no cables or wiring interfaced with the aircraft or other equipment. System arming will be accomplished within the parachute assembly.

f. System shall not interfere with items of crew or aircraft equipment during normal flight operations or emergency egress.

3.3.1.4 Interface with the parachute release fitting. The sensing and actuation package shall directly assemble to the parachute release fitting and thereby become a permanent part of the parachute assembly until the system is fired. At that time the entire package is replaced as a unit. When assembled the unit shall:

a. Not provide additional discomfort or injury to the aircrewman during either normal flight operations or during emergency ground and in-flight egress.

b. Be capable of complete checkout including battery and/or EED replacement, if required, without any disassembly from the release fitting.

c. Not compromise the manual operation of the release fitting with either hand with or without gloves.

d. Be capable of withstanding the normal rough handling of a parachute assembly when not installed in an ejection seat.

e. Not cause any degradation in strength of components which are subject to parachute opening shock. The parachute release in either manual or automatic modes shall be capable of withstanding a static load of 7500 lbs.

f. Not in any way degrade the performance of the escape system.

g. When actuated, lock in the open position to preclude re-connection of the parachute attachment fitting.

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3.3.1.5 Electrical

3.3.1.5.1 Insulation resistance. The insulation resistance between the case and the electrical circuit shall be greater than 100 megaohms.

3.3.1.5.2 Soldering. Electrical connectors shall be mechanically secure and electrically continuous after soldering.

3.3.1.6 Thermal considerations. The Electronics Assembly shall be capable of being cooled by natural convection and radiation without the addition of finned heat sinks, etc. external to the module. Any internal heat dissipating components shall be protected such that no thermal damage to adjacent components will result.

3.3.1.7 Interchangeability and replaceability. Parts and assemblies of all systems shall be interchangeable or replaceable in accordance with MIL-I-8500.

3.3.1.8 Precautions for electrical systems. Electrical systems shall be protected from hazards of electromagnetic radiation (HERO) in accordance with MIL-STD-461 and MIL-STD-1385 and shall be independently shielded circuits secured against vibration and protected against short circuits or grounding resulting from chafing. The electrical power supply shall be adequate to insure ignition under all service environmental conditions specified herein. Specific examples of methods, components, and techniques for implementing precautions are included in NAVWEPS OD 30393. All electrical wiring shall,where required, be an integral part of the harness system installed in the aircraft or worn by the aircrewman.

3.3.1.9 Design safety. The design of the hardware shall conform to the requirements of MIL-STD-882 to ensure system safety.

a. Immunity to rain and humidity. The system, including its sensors, shall not operate prior to water entry if it is deluged by heavy rains, conditions of 100% humidity or during any moisture condition associated with descent through clouds, snow and sleet or any operations taking place in a sea environment.

b. Single failure mode. The following failure modes shall be eliminated/controlled as specified.

(1) No single failure shall result in early release of either parachute canopy release fitting.

(2) Electrical circuits shall be protected from hazards of electromagnet radiation to preclude early firing of EED cartridges.

c. Relays and switches within the systems shall not actuate or fail due to shock loads such as may occur during rough handling.

d. Electro-explosive devices shall not fire as a result of exposure to high ambient temperatures, or as a result of battery failure.

3.3.1.10 Reliability. A reliability program shall be established and maintained in accordance with the requirements of AR-49B as they apply to a component or subsystem of an Aircrew Automated Escape System (AAES). As part of the reliability analysis to

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be submitted, the contractor shall prepare a failure mode analysis for each escape system component and subsystem which could, by failing, adversely affect the survival of the aircrewman. This analysis shall provide (a) the probability of failure, (b) the expected mode(s) of failure, and (c) the consequence of each mode of failure. Appropriate reliability requirements will be imposed on suppliers of purchased parts. In accordance with paragraphs 3.1.4 and 3.3.2, all electrical components are to be standard Established Reliability (ER) parts. Where such parts are not available, MIL-STD or special parts screened to equivalent ER levels may be used. Microcircuits shall comply with MIL-M-38510A for the Class B device.

a. Quantitative Requirement. The reliability of a complete system, consisting of one LH and one RH automatic parachute canopy release fitting, shall be .96 at the 90% confidence level. The reliability of one automatic parachute canopy release fitting shall be .98 under the operational and environmental conditions specified herein, and shall be demonstrated at the 90% confidence level.

No single failure mode shall result in early release of a harness fitting.

b. Inadvertent operation. The reliability of one unit with respect to an inadvertent operation of the automatic release shall be .99 at the 90 percent lower confidence level.

3.3.1.11 Maintainability

a. Maintainability manhour requirements. The maximum acceptable measure of maintainability shall be one manhour of direct maintenance labor during scheduled inspection of the parachute. Repair times shall be computed on the basis of accomplishment of maintenance by personnel with a Navy "PR-2" rating.

b. Maintainability program requirements. The maintainability program to achieve, demonstrate and assure retention of the unit shall be in accordance with MIL-STD-470 and MIL-STD-471.

(1) Compliance with MIL-STD-470 will be limited to a throwaway concept.

(2) Compliance with MIL-STD-471 will be limited to demonstrating: the removal/replacement of the battery(s), removal/replacement of EED, and performing the functional check.

c. Construction and Packaging. In addition to the requirements of Standard MIL-STD-454, the following requirements shall apply.

(1) The unit should be designed/constructed as a non-repairable (throw-away-on-failure).

(2) The packaging/construction of the unit shall provide for easy access, removal and replacement of the battery(s) and EED. Provisions shall also be included to insure that the battery(s) cannot be installed incorrectly.

3.3.2 Detail design requirements. In addition to the general requirements of paragraph 3.3.1, the following detail requirements shall apply.

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3.3.2.1 System requirements. The following requirements refer to the total sensing and actuation package.

3.3.2.1.1 Maintenance concept. The following maintenance concept is established.

a. The unit shall have a scheduled maintenance interval of 210 days and will be accomplished concurrent with the scheduled inspection interval of the parachute when possible. During this inspection, the following actions will be accomplished:

(1) The battery will be checked by extracting energy to exercise the electronic circuit.

(2) A unit functional test will be made to insure operation of the circuitry and continuity of the EED.

(3) A unit test will be made to include a check to assure no inadvertent operation.

(4) A visual check of the unit will be made for damage that would preclude proper operation.

b. The battery will be replaced with a new off-the-shelf battery at a scheduled maintenance interval of not sooner than 2 years.

3.3.2.1.2 Design constraints for maintainability

a. General. The design, configuration, installations and operation of the equipment shall minimize maintenance/support requirements including the need for special tools and support equipment, personnel skills, manpower and elapsed maintenance time, in accordance with MIL-STD-1472.

b. Storage life. The system shall have a minimum of five years storage life (unit is considered to be stored without a battery(s)).

c. Service life. The system shall have a minimum of two years installation service life for all components.

d. Fault isolation and test provisions. The design shall include provisions for making a functional test to insure operation of the circuits.

3.3.2.2 Power Source. The power source shall be self-contained and have no dependence on existing aircraft power.

3.3.2.3 Electro-explosive device. All ballistic components, including but limited to cartridges, cartridge-actuated devices, and ballistic hoses and fittings, shall be approved by the government procuring activity prior to installation or use in the system. All cartridges and cartridge-actuated devices shall conform to the requirements of MIL-D-21625 and MIL-D-23615 and, if applicable, MIL-I-23659. Other components such as chemical, pneumatic, mechanical, etc. shall conform to appropriate standards approved by the government procuring activity.

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3.3.2.3.1 Design requirements

a. The initiator/actuator assembly shall release the canopy release adapter and lock open the automatic canopy release under a constant load of 600 lbs.

b. The EED shall fire when supplied 3.5 amperes at 3.5 volts DC for a time duration of 10 milliseconds maximum.

3.3.2.4 Control electronics. Semiconductor components shall be type JANTX to the extent possible.

3.3.2.5 Microcircuit. The assembly shall be totally enclosed and sealed except for the input/output leads. The microcircuit shall be designed and tested to meet the requirements of MIL-M-38510 Class B. Either dual in-line or flat pack design is acceptable.

3.4 Information for Government-Furnished Equipment. The contractor shall submit to the government procuring activity, as soon as practical, all detail design requirements necessary in the development of the various components of the system to be government furnished. The submittal of this data shall be accomplished in sufficient time for the phasing-in of all Government-furnished equipment.

3.5 Marking and decalcomanias. Decalcomanias used in conjunction with the system installation shall conform to the requirements of MIL-D-8634. General guidance for the application of the decalcomanias is provided in MIL-D-23890. Labeling shall be in accordance with MIL-STD-1472 and MIL-A-25165 and the following requirements.

3.5.1 Preflight check-off list and decalcomanias. The preflight check-off list and any decalcomanias determined to be required should include any preflight checks to be performed on the system to verify that it is functioning properly. Decalcomanias shall be located in a prominent area so as to be readily observed by the aircrewman.

3.6 System/component data requirements. Data shall be supplied in accordance with the applicable Contract Data Requirements List (CDRL).

3.7 Identification of product. Identification marking shall be in accordance with MIL-STD-130. Contents of the nameplate shall be in accordance with MIL-N-18307.

3.8 Workmanship. Workmanship shall be of the highest quality to assure optimum performance, reliability, and service life.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved

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by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Production lot testing principles apply.

4.2 Test requirements. The test programs required herein for the automatic system and its components must be completed prior to system release for operational evaluation. All test articles used in performing the required tests shall conform to the automatic system production design. Any deviation from the production design must be approved by the government procuring activity. Design deviation approval shall be obtained prior to performing any test. Changes in the automatic system and/or component design resulting from the required tests may necessitate retesting using the revised system configuration.

4.2.1 Component/Subsystem qualification tests. Component/subsystem qualification tests shall be conducted in accordance with applicable military specifications and standards.

4.2.2 Environmental test regimen. The environmental testing of the automatic system shall be conducted on one or more systems in accordance with MIL-T-5422 an MIL-STD-810 as identified below and shall be conducted in the order listed below:

Category A	Category B
(1) Temperature-Altitude Test (MIL-T-5422)	(2) Vibration Test (MIL-T-5422)
(3) Shock Test (MIL-T-5422)	
(4) Humidity Test (MIL-T-5422)	(5) Salt Fog Test (MIL-T-5422)
(6) Sand and Dust Test (MIL-T-5422)	
(7) Temperature Shock Test (MIL-T-5422)	(8) Rain (MIL-STD-810)
	(9) Fungus (MIL-T-5422)
(11) Acceleration (± 20 g on all three axes)	(10) Handling Shock (12 random drops from a height of six feet onto a concrete floor)
	(12) Demagnitization

For the salt fog test, the duration of the test shall be 168 hours. Sulphur dioxide gas shall be injected into the chamber at 6 hour intervals. The gas shall be fed in for a period of 60 minutes at a flow rate of 50 cc per hour per cubic foot of chamber volume.

4.2.3 Manual egress regimen. The ejection seat escape system shall be tested to ensure that manual egress by fully equipped 3rd through 98th percentile aircrewman from the seat installed in the applicable air-crew stations, and from the aircraft, is not compromised.

4.2.3.1 Demonstration of aircrew capability to reach and actuate aircrew controls. For each aircrew station the contractor shall demonstrate compliance with the requirements of 3.2.1(b) of MIL-S-18471. The demonstration shall be performed in the actual aircrew station.

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4.2.4 Aircrew escape system component qualification. Aircrew automatic escape system approval for service use requires that:

a. Cartridges and cartridge actuated devices be qualified in accordance with the requirements of MIL-D-21625, MIL-D-23615, and MIL-I-23659, respectively.

b. The personnel parachute/hardware subsystem be qualified as specified by the government procuring activity.

Failure of any or these components during other required tests of the escape system and/or its components may be grounds for revocation of the failed component's qualification and the performance of additional component design and testing to regain component qualification.

4.2.5 Design Feasibility Testing. 30 units will be tested as follows:

10 tests at water temperature 29°F
10 tests at water temperature 57°F
10 tests at water temperature 86°F

4.2.6 Qualification Test Program. Specified tests shall be conducted to ensure that the capability specified herein has been provided, and that all emergency equipment and components will function properly throughout the extreme range of the escape envelope. The following data shall be obtained during the tests:

a. Verification that all automatic subsystems and components functioned properly.

b. Meteorological data, including:

(1) Wind velocity and direction and whether steady, shifting or gusty.

(2) Temperature

(3) Sea State for all over water testing.

Tests should be conducted under meteorological extremes as specified herein. Data acquisition shall commence at the start of the test and shall continue through all phases of the test until approximately 5 minutes after water impact of the test subject and equipment combination. To obtain these data, the test facility may utilize any commonly-accepted instrumentation methods. The test facility shall ensure that sufficient photographic and telemetric coverage is provided to show parachute opening shock forces on all subsystems and components. Data shall be obtained on all events that occur from arming of the system to final recovery of the test subject or dummy.

4.2.6.1 Specific Qualification Test Program Elements. The following are required elements of the qualification test program:

a. Environmental Testing 24 units
(in accordance with 4.2.2)

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(1) Test 12 units by subjecting 2 units to each of the six Category A regimens. Each unit will be checked out on the test set prior to the environmental test, and eight times after the environmental test. The unit will then be rain tested and fired.

(2) Test the remaining 12 units by subjecting each to all the Category A and Category B test regimens in a sequence corresponding to their numerical order. Check each unit on the test set after each regimen. At the completion of all, 12 regiments subject each test to rain spray and then fire it.

(3) After completion of all environmental tests, each unit is to be operationally tested. Inadvertent operation water spray rainwater tests will precede each operational test. The rainwater inadvertent operation tests shall be conducted in accordance with Method number 506.1 of MIL-STD-810 for steady state rain conditions for a time period of 15 minutes in a vertical orientation simulating flight conditions with water of 1000 micromhos conductivity.

b. Water Drops (8 tests) 16 units

(1) Four tests with inflated life preserver.

(2) Four tests with uninflated life preserver.

c. HERO Tests 2 units

d. Ejection Tests 6 units. Six units shall be subjected to full scale ejection tests. At the completion of all ejection tests, the units shall be checked out on the test set, rain spray tested, and fired.

e. Parachute Drops (21 tests) 42 units

(1) 18 Drops/Live Jumps at wind speeds up to 20 knots and up to sea state 6.

(2) 3 Drops at wind speed greater than 30 knots and sea state greater than 6.

(3) Live test parachutists shall be used in preference to dummies whenever test conditions allow the use of live subjects.

(4) At least 50 percent of all parachute drop testing shall be performed into the ocean.

4.2.6.2 Refurbishment of Test Items. Refurbished units may be used to minimize test hardware costs. Refurbishment techniques or errors shall not be considered a valid rationale for declaration of a "no-test" condition if failure of a refurbished unit should occur.

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4.2.7 Reliability Demonstration. The tests specified in paragraph 4.2.6.1a(1) may be used to demonstrate the reliability requirements of paragraph 3.3.1.10. Additional test articles from the qualification test program may be used, if required, to satisfy the reliability demonstration requirements.

4.3 Quality assurance program for acceptance of deliveries. The provisions of AD42 will be strictly adhered to.

4.3.1 Non-destructive testing of critical parts. The contractor shall identify those system parts/components which, should they structurally fail during escape system operation or during application of crash loads would result in system failure likely to be injurious to aircrewmen. The contractor shall recommend to the procuring activity a program of non-destructive testing for assuring acceptable production quality of the identified critical parts/components.

4.3.2 Inspection of production units. Production systems shall satisfactorily complete a non-destructive functional test program prior to acceptance under contract. The program shall be developed by the prime contractor and submitted via the Administrative Contract Officer and the activities designated within the contract to the procuring activity for approval. The test program shall consist of sufficient functional tests to demonstrate overall system performance. Production lot tests will be conducted by a designated government or independent testing laboratory.

Preparing Activity
Navy - As

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DOCUMENT IDENTIFIER AND TITLE: System, Automatic, Water Activated Parachute Release MIL-S-85076(AS)

NAME OF ORGANIZATION AND ADDRESS

Naval Air Systems Command (AIR-5312)
Washington, D.C. 20361

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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